PROBABILISTIC REASONING FOR STREAMING ANOMALY DETECTION

Kevin M. Carter and William W. Streilein

MIT Lincoln Laboratory, 244 Wood St., Lexington, MA 02420
{kevin.carter, wws}@ll.mit.edu

ABSTRACT

In many applications it is necessary to determine whether an observation from an incoming high-volume data stream matches expectations or is anomalous. A common method for performing this task is to use an Exponentially Weighted Moving Average (EWMA), which smooths out the minor variations of the data stream. While EWMA is efficient at processing high-rate streams, it can be very volatile to abrupt transient changes in the data, losing utility for appropriately detecting anomalies. In this paper we present a probabilistic approach to EWMA which dynamically adapts the weighting based on the observation probability. This results in robustness to data anomalies yet quick adaptability to distributional data shifts.

***Index Terms*—** Anomaly detection, predictive models, time series analysis, statistical learning, information security

1. INTRODUCTION

The field of *streaming analysis*, in which analytics get a single pass at the data, is gaining usage in analysis of network traffic, stock trading, search queries, and other high-volume, real-time applications. Systems that analyze data streams often have a large volume of memory, minimal storage, and require low computational complexity. Given the real-time nature of data streams, they are often utilized towards the analysis of time-series data; typically for filtering, aggregation, prediction, characterization, and anomaly detection.

Consider the case in which an anomaly detection system is monitoring numerous – tens of thousands or more – unique 1-dimensional signals received from a sensor aggregating data over some time window of length T. For example, in network traffic analysis, these signals may consist of the number of bytes sent by unique IP addresses within a 10 minute window[1]. While raw data is plentiful, we assume that labeled training data, detailing anomalous activity, is unavailable. In this scenario, it is desirable to have a single behavioral model that can be applied to multiple signals in which the model learns the parameterization from the signal itself and can detect anomalous activity.

For a thorough survey of anomaly detection work, please refer to [2], which offers definitions of different types of anomalies, both spatial and temporal, and describes much of the work in space. Spatial and classification based approaches are irrelevant for the task of anomaly detection on streaming time-series data. Several methods have leveraged autoregressive (integrated) moving average techniques for both univariate and multivariate time series [2], although many of these methods have not considered the streaming analysis case for large volume signals.

In the field of process control, the Exponentially Weighted Moving Average (EWMA) [3] is a popular algorithm for monitoring the state of a physical process to determine if its activity, as reflected in some measured performance variable, is within expected limits [4]. Prior enhancements to EWMA designed to react quickly to abrupt yet permanent changes in process behavior [5, 6] are intentionally not robust to anomalies, while methods designed to be robust to outliers are typically rule-based [4] and do not adjust well when the process undergoes an abrupt distributional shift.

We present a probabilistic method of EWMA which dynamically adjusts the parameterization based on the probability of the given observation. This method produces dynamic, data-driven anomaly thresholds which are robust to abrupt transient changes, yet quickly adjust to long-term distributional shifts. Our method has minimal complexity and requires nearly no *a priori* parameterization, scaling to numerous signals without providing unique consideration to individuals. We refer to this methodology as Probabilistic EWMA (PEWMA) and will show that the anomaly thresholds provided are tighter and better adapted to the data stream than prior EWMA methods.

2. ANOMALY DETECTION WITH EWMA

For the purposes of this paper, we define an anomaly as an observation in a temporal signal that is statistically unlikely given the generative distribution. We distinguish between three types of signal changes as illustrated in Fig. 1 (from left to right): an abrupt transient change that is short lived, an abrupt shift to the underlying signal distributional which is

This work is sponsored by the Department of Defense under Air Force Contract FA8721-05-C-0002. Opinions, interpretations, conclusions and recommendations are those of the author and are not necessarily endorsed by the United States Government.

978-1-4673-0183-1/12/$31.00 ©2012 IEEE

Authorized licensed use limited to: MINCYT. Downloaded on October 22,2026 at 17:26:56 UTC from IEEE Xplore. Restrictions apply.

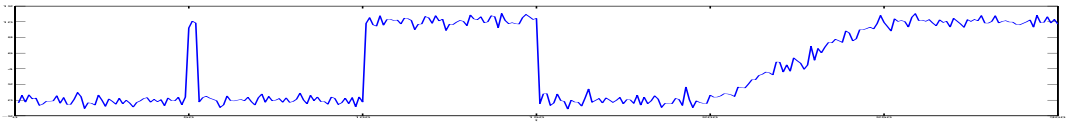


Fig. 1: Three types of major signal changes: abrupt transient, abrupt distributional shift, and gradual distributional shift

long lived, and a gradual shift to the distribution, eventually resulting in a new signal state. For the anomaly detection task, it is desirable to alert on all abrupt transient changes[1], initially alert on abrupt distributional shifts but have the model quickly adjust to the change, and not alert on gradual shifts. We note that this definition does not cover all anomalous activity, such as a gradual increase in temperature of a refrigerator, but we will focus on the class of abrupt anomalies described.

The Exponentially Weighted Moving Average (EWMA) [3] computes the local mean μ_t of a time series $\{X_t\}$ by applying exponentially decreasing weight factors to each prior observation, such that $\mu_t = \sum_{i=1}^{t} \alpha^{t-i}(1-\alpha)X_i, 0 \leq \alpha < 1$. The parameter α specifies the amount of weight to put on historical values vs. the most recent samples. This is typically calculated using an update equation:

$$\mu_t = \alpha\mu_{t-1} + (1-\alpha)X_t, \tag{1}$$

which makes the EWMA very efficient for online analysis of large data streams. Note that this equation is a form of first-order autoregressive modeling. For the purposes of anomaly detection, it is generally applicable to alert on any $|X_t - \mu_t| > m\sigma_t$, where m is a constant multiplier and σ_t is the estimated standard deviation of $\{X_t\}$ at time t.

The use of EWMA implies an inherent assumption that the sample mean, while potentially non-stationary, is changing only gradually with respect to the parameterization, and is therefore very susceptible to abrupt anomalous events. Equation (1) shows that even as α nears 1, a significant change in X_t will result in a significant increase in μ_t, and an even greater impact on σ_t. The resultant model will have a poorer fit to the data, lessening the detection capabilities.

3. PROBABILISTIC EWMA

To overcome the shortcomings of EWMA, we deploy a scheme which adjusts the weighting parameter α in (1) based on the probability of the current sample. Let $\{X_t\}$ be the signal of measured observations through time t. We are able to estimate the local mean of the process, μ_t, as the exponentially and probabilistically weighted moving average of the previous samples:

$$\mu_t = \alpha(1-\beta P_t)\mu_{t-1} + (1-\alpha(1-\beta P_t))X_t, \tag{2}$$

where $0 < \alpha < 1$ is the maximal weighting parameter, P_t is the probability X_t evaluated under some modeled distribution, and β is the weight placed on P_t. Note that as $\beta \to 0$, PEWMA converges to a standard EWMA. We choose to adapt weights α by $1 - \beta P_t$ such that samples that are less likely to have been observed offer little influence to the updated estimate. While other monotonic transformations may be explored, our linear scheme is simple and has the appropriate effects.

[1]This does not include periodic shifts which can be predicted

Algorithm 1 Probabilistic EWMA

Require: $X_t, \hat{X}_t, \hat{\sigma}_t, T, t$
Ensure: $P_t, \hat{X}_{t+1}, \hat{\sigma}_{t+1}$
  $Z_t \leftarrow \frac{X_t - \hat{X}_t}{\hat{\sigma}_t}$
  $P_t \leftarrow \frac{1}{\sqrt{2\pi}} \exp(-\frac{Z_t^2}{2})$
  if $t \leq T$ **then**
    $\alpha_t \leftarrow 1 - 1/t$
  else
    $\alpha_t \leftarrow (1 - \beta P_t)\alpha$
  end if
  $s_1 \leftarrow \alpha_t s_1 + (1-\alpha_t)X_t$
  $s_2 \leftarrow \alpha_t s_2 + (1-\alpha_t)X_t^2$
  $\hat{X}_{t+1} \leftarrow s_1$
  $\hat{\sigma}_{t+1} \leftarrow \sqrt{s_2 - s_1^2}$

In order to use Eq. (2) for prediction purposes, we operate under the assumption that locally $\{X_t\}$ is wide-sense stationary, hence the estimated mean at the current time is the minimum mean square error estimate for the value at the next time iteration. We are able to estimate a local standard deviation $\hat{\sigma}_{t+1}$ as described in Algorithm 1, and use this parameter both for the evaluation of P_t and for setting detection thresholds.

For the purposes of this paper, let us make a Gaussian signal assumption and compute the probability, normalized such that $0 < P_t < \frac{1}{\sqrt{2\pi}}$. Specifically, letting $\hat{X}_t = \mu_{t-1}$ be the predicted value and $\hat{\sigma}_t$ represent the standard deviation at time t, we normalize the observation to a zero-mean and unit standard deviation random variable, $Z_t = (X_t - \hat{X}_t)/\hat{\sigma}_t$, and compare to a standard normal distribution, $P_t = \frac{1}{\sqrt{2\pi}} \exp(-\frac{Z_t^2}{2})$, to obtain our probability estimate. This normalization is critical to account for potentially large data scales, where the typical $1/\sqrt{2\pi\sigma^2}$ term causes $P_t \to 0$ as $\sigma \to \infty$, regardless of the value of X_t.

Algorithm 1 presents the steady-state version of PEWMA. In the parlance of [2], we utilize PEWMA to detect point anomalies and contextual anomalies, for which context is de-

Authorized licensed use limited to: MINCYT. Downloaded on October 22,2026 at 17:26:56 UTC from IEEE Xplore. Restrictions apply.

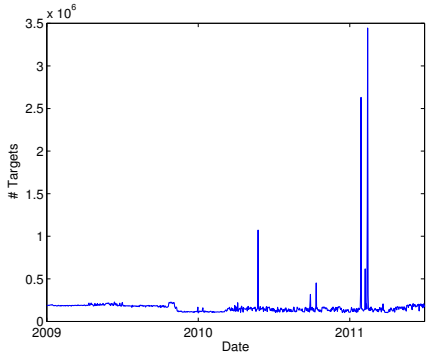


Fig. 2: Daily reported targets of malicious cyber activities from the SANS Internet Storm Center, 01/2009-06/2011.

fined by recent observations. Periodic bursts, for example, *will* be detected by this method and considered false alarms.

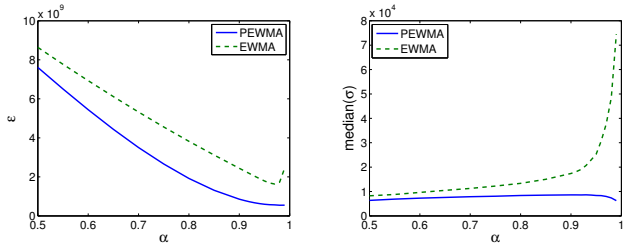
Initialization and Training

In order to operate in a steady state, it is necessary to first train the model on the initial observations during start-up. For these purposes, we will initialize the model parameters according to the first received sample, $s_1 = X_1$ and $s_2 = X_1^2$, and predicting the next value $\hat{X}_2 = X_1$.

It is important to note that EWMA methods can suffer bias towards the initialization point, as it takes a long time to filter out the effects of an anomalous observation at the start up. This can be overcome with a training period T in which the model parameters are set to compute the exact mean and standard deviation of the data. This can still be accomplished by setting $\alpha_t = (1 - 1/t)$ for $t \leq T$.

4. NETWORK SECURITY ANOMALY DETECTION

To demonstrate the improved performance of PEWMA over standard EWMA, we model the predicted number of reported global intrusions in network security, a data set provided by the SANS Internet Storm Center [7]. SANS receives reports of intrusions and malicious activity from across the world, and aggregates these totals into a daily report detailing the volume of these incidents. Anomaly detection is important in these situations, as a significant burst in reporting may be indicative of a large-scale attack. As seen in Fig. 2, the number of reported targets of malicious activity contains abrupt shifts in both the transient and distributional senses. While part of our motivation is to apply a single method to numerous signals, for simplicity of illustration we utilize this signal as a proof-of-concept. Our goal in this simulation is to model the signal and predict the number of targets reporting malicious activity each day, raising an anomaly when this value is outside of expected threshold. While obvious, it is important to note that we are doing this in a causal fashion, in which our predictions are based solely on the past observations.



(a) Prediction MSE (b) Median Standard deviation

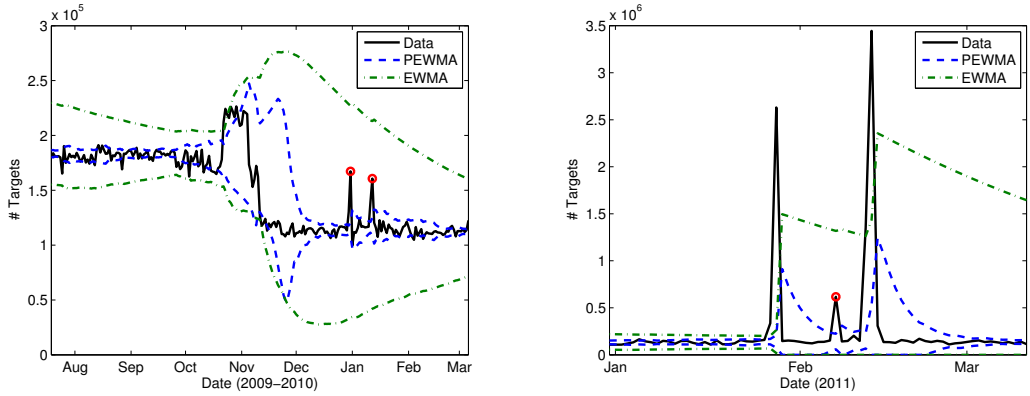
Fig. 3: Comparing PEWMA to standard EWMA, over varying α, in modeling number of daily targets reported to SANS

Anomaly detection systems rely on their ability to predict the data stream. In Fig. 3a we illustrate a comparison between PEWMA and EWMA, plotting the prediction mean-squared error (MSE) per day, varied by the value of α. It is clear that for each value of α, PEWMA outperforms standard EWMA. This is intuitive, as the PEWMA model incorporates probabilistic weighting and is more robust to spurious anomalies. For this experiment, we used $T = 14$ days as our training period for both methods, and $\beta = 1$ for PEWMA. Note that while we computed moving averages on the full data set, we did not include the 5 significant anomalies in the MSE calculation, which skewed the parameterization to favor significantly smaller values of α; PEWMA still outperformed EWMA for all α, however.

We accentuate the increased performance by showing that PEWMA better predicts the data with a lower standard deviation in Fig. 3b. Specifically, this plot illustrates that for varying values of α, PEWMA is relatively consistent with its estimation of the model deviation, while standard EWMA shows very high model deviations as $\alpha \to 1$. In comparison to Fig. 3a, we see that in order to minimize the MSE, it is necessary to increase the variability in EWMA's resultant prediction. On the contrary, PEWMA actually shows reduced variability as $\alpha \to 1$. This becomes critical in the anomaly detection scheme, as the PEWMA model will be able to detect more granular anomalies based on the needs of the user.

While we have shown that PEWMA outperforms standard EWMA for prediction purposes, our we now illustrate our main goal of anomaly detection,. In Fig. 4 we present two specific date ranges which demonstrate the effects of both transient and sustained distributional shifts on the detection systems. Specifically, the system issues an alert whenever the calculated probability $P_t < \tau$, for some threshold τ. This is equivalent to alerting whenever the observation is greater than some number of standard deviations away from the predicted value. For example, setting $\tau = 0.0044$ will alert whenever the observation is in excess of 3 standard deviations from the predicted value. For this test, we used the parameters $\alpha_{EWMA} = 0.97$ and $\alpha_{PEWMA} = 0.99$, which were the optimal minimum MSE parameters based on Fig. 3a.

Authorized licensed use limited to: MINCYT. Downloaded on October 22,2026 at 17:26:56 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 4: Predictive modeling used to set thresholds for anomaly detection. Standard EWMA suffers greatly from the abrupt bursts, missing the blatant anomalies (red circles), while PEWMA quickly recovers and detects.

In Fig. 4a, there are distributional shifts in both mid-Oct. and early-Nov., 2009. Standard EWMA reacts quickly to the first shift, but has the adverse effect of inflating the estimate of the standard deviation. This results in slowly adjusting thresholds during the second shift, yielding a system unable to detect anomalies for an extended duration. While PEWMA was slower to react to the initial shift, due to the low probability of the abrupt change, the prediction bounds were more stable, resulting in system that better adjusted to future changes in the data. If a faster adjustment to the initial shift is desirable, simply lowering α will suffice. In Fig. 4b, this impact is magnified during the anomalous activity between late-Jan. and mid-Feb., 2011. In this scenario, EWMA completely misses the anomaly signaled by the red circle due to the inflated thresholds; meanwhile each visible anomaly is easily detected by the PEWMA model, and the bounds quickly regress to the norm. One can notice the same effect in Fig. 4a.

5. CONCLUSION

In this paper we employed probabilistic weighting to the standard EWMA form, biasing samples that are more likely to have come from the generative distribution of the observed data. Our method is robust to anomalous fluctuations in the data stream which are often prevalent in many practical applications. While the PEWMA model is robust to anomalies, it also shows the ability to adapt to changes in the generative model of the data, and the trade off between this robustness and dynamism is tunable via the model parameters. We have illustrated the utility of PEWMA on an application to anomaly detection in network security and shown how this straight-forward modification to a popular algorithm produces superior results.

In future work we aim to expand the uses of PEWMA to account for contextual anomalies in non-stationary data streams, specifically those with a diurnal cycle to them, such as aggregate network traffic [1]. We also look to reduce the variability of the standard deviation estimate after anomalous events. While PEWMA has shown significant improvement over EWMA in this respect, there is still a need to reduce these potential explosions. Finally, we will look towards second-order modeling to detect slow growing anomalies, as opposed to the abrupt changes we focused on here.

6. REFERENCES

[1] W. Streilein, J. Truelove, C. Meiners, and G. Eakman, "Cyber situational awareness through operational streaming analysis," in *Proceedings of 30th MILCOM*, 2011.

[2] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–72, July 2009.

[3] S. W. Roberts, "Control chart tests based on geometric moving averages," *Technometrics*, vol. 1, no. 3, pp. 239–250, Aug. 1959.

[4] J. M. Lucas and M. S. Saccucci, "Exponentially weighted moving average control schemes: Properties and enhancements," *Technometrics*, vol. 32, no. 1, pp. 1–12, Feb. 1990.

[5] R.-S. Guo and J.-J. Chen, "An EWMA-based process mean estimator with dynamic tuning capabiltiy," *IIE Transaction*, vol. 34, no. 6, pp. 573–582, 2002.

[6] G. S. Wasserman, "An adaptation of the EWMA chart for short run SPC," *International Journal of Production Research*, vol. 33, no. 10, pp. 2821–2833, 1995.

[7] "SANS Internet Storm Center," http://isc.sans.org/reports.html.

Authorized licensed use limited to: MINCYT. Downloaded on October 22,2026 at 17:26:56 UTC from IEEE Xplore. Restrictions apply.